

Monitoring, Optimization & Serverless

Module 6 — AWS Technical Essentials

📈 CloudWatch & CloudTrail

Monitoring and observability for AWS resources and applications.

- Collects metrics from all AWS services
- Custom metrics from your applications
- Dashboards for real-time visualization
- Logs aggregation and search

API audit trail — records every API call in your account.

- Who did what, when, from where
- Enabled by default (90-day history)

🔔 CloudWatch Alarms

- Trigger on metric thresholds
- Actions: SNS notification, Auto Scaling, EC2 action
- States: OK, ALARM, INSUFFICIENT_DATA
- Example: CPU > 80% for 5 minutes

Key pattern: CloudWatch Alarm → SNS Topic → email / Lambda / Auto Scaling

⚖️ Elastic Load Balancing

Distribute traffic across multiple targets for availability and performance.

- Health checks remove unhealthy targets
- Spans multiple AZs and registers Auto Scaling instances automatically

Application LB (ALB)

- Layer 7 (HTTP/HTTPS)
- Path and host-based routing
- Web apps and microservices — **used in Lab 6**

Network LB (NLB)

- Layer 4 (TCP/UDP)
- Ultra-low latency, static IP
- Gaming, IoT, high throughput

📈 EC2 Auto Scaling

Automatically adjust the number of EC2 instances based on demand.

- Set minimum, desired, and maximum capacity
- Scale out on high demand, scale in when idle
- Uses launch templates for instance config
- Spans multiple AZs for high availability

The HA pattern: ELB distributes traffic → Auto Scaling adjusts capacity → CloudWatch triggers scaling

- ELB health checks drive instance replacement
- New instances auto-register with ELB
- Multi-AZ for fault tolerance

⚡ Serverless on AWS

No provisioning, no patching, no scaling decisions. Pay per use.

- **Compute:** Lambda, Fargate
- **Storage:** S3
- **Database:** DynamoDB, Aurora Serverless
- **API:** API Gateway

Pattern: Client → API Gateway → Lambda → DynamoDB

- API Gateway handles HTTP routing and auth
- Lambda runs business logic per request
- DynamoDB stores and retrieves data
- All fully managed, scales to zero

🛠️ Lab Time

Configure High Availability

⏱️ 50 minutes

- Create an Application Load Balancer
- Configure an Auto Scaling group
- Test scaling and fault tolerance